COMPUTER ENGINEERING

QUEZON CITY

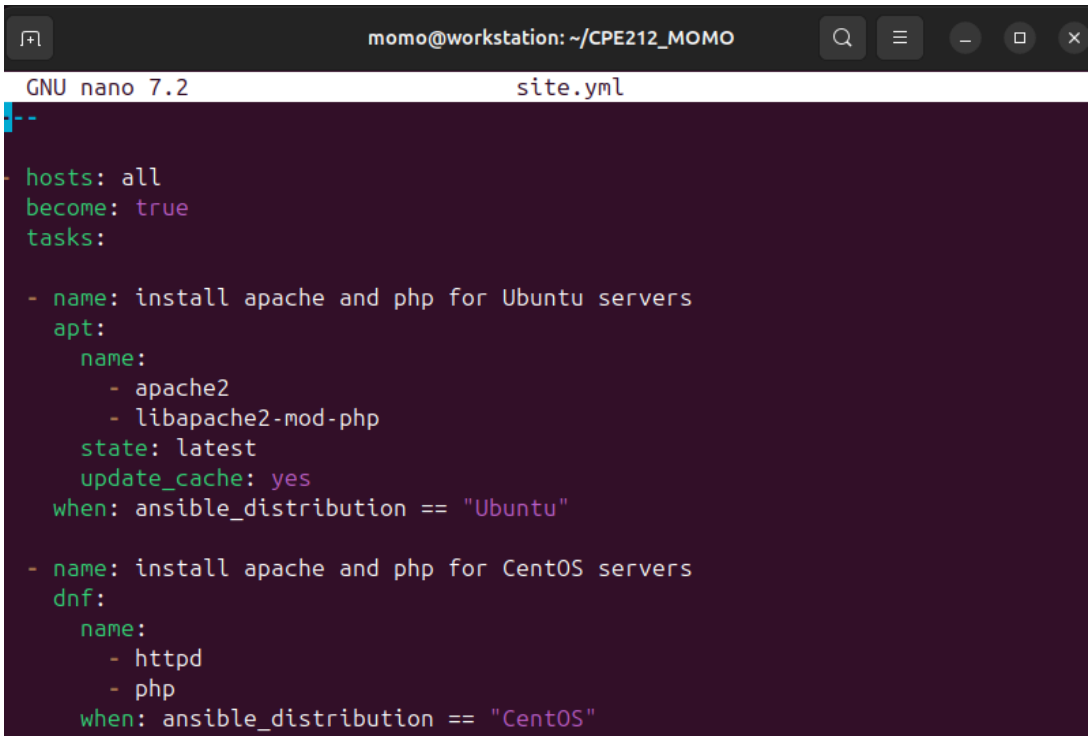
COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/2/2026	**DATE SUBMITTED:**	9/3/2026
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ACTIVITY NO. 6.1 **Targeting Specific Nodes**

1. PROCEDURE OUTPUT

Task 1: Targeting Specific Nodes

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.



2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:

momo@workstation: ~/CPE212_M

GNU nano 7.2 inventory.ini

```
[web_servers]
192.168.56.115
192.168.56.114

[db_servers]
192.168.56.116

[file_servers]
```

Make sure to save the file and exit.

Right now, we have created groups in our inventory file and put each server in its own group. In other cases, you can have a server be a member of multiple groups, for example you have a test server that is also a web server.

3. Edit the *site.yml* by following the image below:

momo@workstation: ~/CPE212_MOMO

GNU nano 7.2 site.yml

```
---

- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

- hosts:  web_servers
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
```

```
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  dnf:
    name:
      - httpd
      - php
    when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

The *pre-tasks* command tells the ansible to run it before any other thing. In the *pre-tasks*, CentOS will install updates while Ubuntu will upgrade its distribution package. This will run before running the second play, which is targeted at *web_servers*. In the second play, apache and php will be installed on both Ubuntu servers and CentOS servers.

Run the *site.yml* file and describe the result.

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --ask-become-pass
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (CentOS)] ***********************************************************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (Ubuntu)] ***********************************************************************************
skipping: [192.168.56.116]
ok: [192.168.56.114]
ok: [192.168.56.115]

PLAY [web_servers] ************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [192.168.56.114]

TASK [install apache and php for Ubuntu servers] ******************************************************************
ok: [192.168.56.114]

TASK [install apache and php for CentOS servers] ******************************************************************
fatal: [192.168.56.114]: FAILED! => {"ansible_facts": {"pkg_mgr": "apt"}, "changed": false, "msg": ["Could not detec
t which major revision of dnf is in use, which is required to determine module backend.", "You should manually speci
fy use_backend to tell the module whether to use the dnf4 or dnf5 backend})"]}

PLAY RECAP ********************************************************************************************************
192.168.56.114             : ok=4    changed=0    unreachable=0    failed=1    skipped=1    rescued=0    ignored=0

192.168.56.115             : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

192.168.56.116             : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- The playbook successfully updated the servers and installed Apache and PHP on the web servers.

4. Let's try to edit again the *site.yml* file. This time, we are going to add plays targeting the other servers. This time we target the *db_servers* by adding it on the current *site.yml*. Below is an example: (Note add this at the end of the playbooks from task 1.3.

```
- name: install mariadb package (CentOS)
  yum:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: "Mariadb- Restarting/ Enabling"
  service:
    name: mariadb
    state: restarted
    enabled: true

- name: install mariadb package (Ubuntu)
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"
```

Make sure to save the file and exit.

Run the *site.yml* file and describe the result.

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --ask-become-pass
BECOME password:

PLAY [db_servers] ***************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.115]
ok: [192.168.56.116]

TASK [install mariadb package (CentOS)] *****************************************************************************
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [install mariadb package (Ubuntu)] *****************************************************************************
skipping: [192.168.56.116]
changed: [192.168.56.115]

TASK [Mariadb- Restarting/ Enabling] ********************************************************************************
changed: [192.168.56.115]
changed: [192.168.56.116]

PLAY RECAP **********************************************************************************************************
192.168.56.115             : ok=3    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

192.168.56.116             : ok=3    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0


momo@workstation:~/CPE212_MOMO$
```

- The playbook successfully targeted the database servers and installed MariaDB on them.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: *systemctl status mariadb.* Do this on the CentOS server also.

```
[momo@centos ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
     Active: active (running) since Thu 2026-09-03 09:30:36 PST; 1min 25s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 12746 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited,>
    Process: 12769 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.ser>
    Process: 12822 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exite>
   Main PID: 12804 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 9 (limit: 22978)
     Memory: 65.3M (peak: 67.5M)
        CPU: 531ms
     CGroup: /system.slice/mariadb.service
             └─12804 /usr/libexec/mariadbd --basedir=/usr

Sep 03 09:30:36 centos systemd[1]: Starting MariaDB 10.5 database server...
Sep 03 09:30:36 centos mariadb-prepare-db-dir[12769]: Database MariaDB is proba>
Sep 03 09:30:36 centos mariadb-prepare-db-dir[12769]: If this is not the case, >
Sep 03 09:30:36 centos systemd[1]: Started MariaDB 10.5 database server.
```

```
momo@server1:~/Desktop$ systemctl status mariadb
● mariadb.service - MariaDB 10.11.14 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
     Active: active (running) since Thu 2026-09-03 01:30:36 UTC; 2min 12s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 10745 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /va>
    Process: 10747 ExecStartPre=/bin/sh -c systemctl unset-environment _WSREP_S>
    Process: 10749 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] &&>
    Process: 10822 ExecStartPost=/bin/sh -c systemctl unset-environment _WSREP_>
    Process: 10825 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0>
   Main PID: 10809 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 11 (limit: 30339)
     Memory: 78.6M (peak: 82.6M)
        CPU: 660ms
     CGroup: /system.slice/mariadb.service
             └─10809 /usr/sbin/mariadbd

Sep 03 01:30:36 server1 mariadbd[10809]: 2026-09-03  1:30:36 0 [Note] InnoDB: L>
Sep 03 01:30:36 server1 mariadbd[10809]: 2026-09-03  1:30:36 0 [Note] Plugin 'F>
Sep 03 01:30:36 server1 mariadbd[10809]: 2026-09-03  1:30:36 0 [Warning] You ne>
```

Describe the output.

- The output shows that MariaDB is installed and running properly on the Ubuntu and CentOS servers.

6. Edit the *site.yml* again. This time we will append the code to configure installation on the *file_servers* group. We can add the following on our file.

```
  GNU nano 7.2
---

- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    package:
      name: samba
      state: latest
```

Make sure to save the file and exit.

Run the *site.yml* file and describe the result.

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --ask-become-pass
BECOME password:

PLAY [file_servers] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.114]

TASK [install samba package] ***************************************************************************************
ok: [192.168.56.114]

PLAY RECAP *********************************************************************************************************
192.168.56.114             : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

- The playbook successfully targeted the file server and installed the Samba package.
  The testing of the *file_servers* is beyond the scope of this activity, and as well as our topics and objectives. However, in this activity we were able to show that we can target hosts or servers using grouping in ansible playbooks.

Task 2: Using Tags in running playbooks

In this task, our goal is to add metadata to our plays so that we can only run the plays that we want to run, and not all the plays in our playbook.

1. Edit the *site.yml* file. Add tags to the playbook. After the name, we can place the tags: *name_of_tag*. This is an arbitrary command, which means you can use any name for a tag.

```
---

- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    tags: always
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    tags: always
    apt:
      upgrade: dist
      update_cache: yes
    when: ansible_distribution == "Ubuntu"
```

```
- hosts: web_servers
  become: true
  tasks:

  - name: install apache and php for Ubuntu servers
    tags: apache,apache2,ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
    when: ansible_distribution == "Ubuntu"

  - name: install apache and php for CentOS servers
    tags: apache,centos,httpd
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

```
- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    tags: centos, db,mariadb
    dnf:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb packege (Ubuntu)
    tags: db, mariadb,ubuntu
    apt:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"

- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    tags: samba
    package:
      name: samba
      state: latest
```

Make sure to save the file and exit.
Run the *site.yml* file and describe the result.

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (CentOS)] ************************************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (Ubuntu)] ************************************************************
skipping: [192.168.56.116]
changed: [192.168.56.115]
changed: [192.168.56.114]

PLAY [web_servers] *************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.115]
ok: [192.168.56.114]
ok: [192.168.56.116]

TASK [install apache and php for ubuntu servers] *******************************************
skipping: [192.168.56.116]
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [install apache and php for CentOS servers] *******************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

PLAY [db_servers] **************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.116]
ok: [192.168.56.114]

TASK [install mariadb package (CentOS)] ****************************************************
skipping: [192.168.56.114]
ok: [192.168.56.116]

TASK [install mariadb packege (Ubuntu)] ****************************************************
skipping: [192.168.56.116]
changed: [192.168.56.114]

TASK [Mariadb-Restarting/Enabling] *********************************************************
changed: [192.168.56.114]
changed: [192.168.56.116]

PLAY [file_servers] ************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.114]

TASK [install samba package] ***************************************************************
ok: [192.168.56.114]
```

```
PLAY RECAP *********************************************************************************************************
192.168.56.114             : ok=9    changed=3    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

192.168.56.115             : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

192.168.56.116             : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0


momo@workstation:~/CPE212_MOMO$
```

- The output shows the available tags that can be used to run specific tasks in the playbook.

2. On the local machine, try to issue the following commands and describe each result:

2.1 *ansible-playbook --list-tags site.yml*

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook --list-tags site.yml

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
momo@workstation:~/CPE212_MOMO$
```

2.2 *ansible-playbook --tags centos --ask-become-pass site.yml*

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook --tags centos --ask-become-pass site.yml
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.115]
ok: [192.168.56.116]
ok: [192.168.56.114]

TASK [install updates (CentOS)] ************************************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (Ubuntu)] ************************************************************
skipping: [192.168.56.116]
changed: [192.168.56.115]
changed: [192.168.56.114]

PLAY [web_servers] *************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [192.168.56.114]
ok: [192.168.56.116]
ok: [192.168.56.115]

TASK [install apache and php for CentOS servers] *******************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]
```

```
PLAY [db_servers] ***************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.116]
ok: [192.168.56.114]

TASK [install mariadb package (CentOS)] *****************************************************************************
skipping: [192.168.56.114]
ok: [192.168.56.116]

PLAY [file_servers] *************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.114]

PLAY RECAP **********************************************************************************************************
192.168.56.114             : ok=5    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

192.168.56.115             : ok=3    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

192.168.56.116             : ok=6    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0


momo@workstation:~/CPE212_MOMO$
```

2.3 *ansible-playbook --tags db --ask-become-pass site.yml*

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook --tags db --ask-become-pass site.yml
BECOME password:

PLAY [all] **********************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.115]
ok: [192.168.56.116]
ok: [192.168.56.114]

TASK [install updates (CentOS)] *************************************************************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (Ubuntu)] *************************************************************************************
skipping: [192.168.56.116]
changed: [192.168.56.114]
changed: [192.168.56.115]

PLAY [web_servers] **************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.115]
ok: [192.168.56.116]
ok: [192.168.56.114]
```

```
PLAY [db_servers] ***************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.116]

TASK [install mariadb package (CentOS)] *****************************************************************************
skipping: [192.168.56.114]
ok: [192.168.56.116]

TASK [install mariadb packege (Ubuntu)] *****************************************************************************
skipping: [192.168.56.116]
ok: [192.168.56.114]

PLAY [file_servers] *************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.114]

PLAY RECAP **********************************************************************************************************
192.168.56.114             : ok=6    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

192.168.56.115             : ok=3    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

192.168.56.116             : ok=5    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0


momo@workstation:~/CPE212_MOMO$
```

2.4 *ansible-playbook --tags apache --ask-become-pass site.yml*

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook --tags apache --ask-become-pass site.yml
BECOME password:

PLAY [all] **********************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (CentOS)] *************************************************************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (Ubuntu)] *************************************************************************************
skipping: [192.168.56.116]
changed: [192.168.56.114]
changed: [192.168.56.115]

PLAY [web_servers] **************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.115]
ok: [192.168.56.116]
ok: [192.168.56.114]
```

```
TASK [install apache and php for ubuntu servers] ********************************************************
skipping: [192.168.56.116]
ok: [192.168.56.114]
ok: [192.168.56.115]

TASK [install apache and php for CentOS servers] ********************************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

PLAY [db_servers] ***************************************************************************************

TASK [Gathering Facts] **********************************************************************************
ok: [192.168.56.116]
ok: [192.168.56.114]

PLAY [file_servers] *************************************************************************************

TASK [Gathering Facts] **********************************************************************************
ok: [192.168.56.114]

PLAY RECAP **********************************************************************************************
192.168.56.114             : ok=6    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

192.168.56.115             : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

192.168.56.116             : ok=5    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0


momo@workstation:~/CPE212_MOMO$
```

2.5 *ansible-playbook --tags "apache,db" --ask-become-pass site.yml*

```
momo@workstation:~/CPE212_MOMO$ ansible-playbook --tags "apache,db" --ask-become-pass site.yml
BECOME password:

PLAY [all] **********************************************************************************************

TASK [Gathering Facts] **********************************************************************************
ok: [192.168.56.115]
ok: [192.168.56.114]
ok: [192.168.56.116]

TASK [install updates (CentOS)] *************************************************************************
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

TASK [install updates (Ubuntu)] *************************************************************************
skipping: [192.168.56.116]
changed: [192.168.56.114]
changed: [192.168.56.115]

PLAY [web_servers] **************************************************************************************

TASK [Gathering Facts] **********************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.115]
ok: [192.168.56.116]
```

```
skipping: [192.168.56.114]
skipping: [192.168.56.115]
ok: [192.168.56.116]

PLAY [db_servers] ***************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.114]
ok: [192.168.56.116]

TASK [install mariadb package (CentOS)] *****************************************************************************
skipping: [192.168.56.114]
ok: [192.168.56.116]

TASK [install mariadb packege (Ubuntu)] *****************************************************************************
skipping: [192.168.56.116]
ok: [192.168.56.114]

PLAY [file_servers] *************************************************************************************************

TASK [Gathering Facts] **********************************************************************************************
ok: [192.168.56.114]

PLAY RECAP **********************************************************************************************************
192.168.56.114             : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

192.168.56.115             : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

192.168.56.116             : ok=6    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```

Task 3: Managing Services

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.

```
- name: install apache and php for CentOS servers
  tags: apache,centos,httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache, centos,httpd
  service:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"
```

Figure 3.1.1

Make sure to save the file and exit.

```
- name: install apache and php for CentOS servers
  tags: apache,centos,httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"
```

You would also notice from our previous activity that we already created a module that runs a service.

```
- hosts: db_servers
  become: true
  tasks:

  - name: install mariadb package (CentOS)
    tags: centos, db,mariadb
    dnf:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true
```

Figure 3.1.2

```
- name: install mariadb packege (Ubuntu)
  tags: db, mariadb, ubuntu
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"

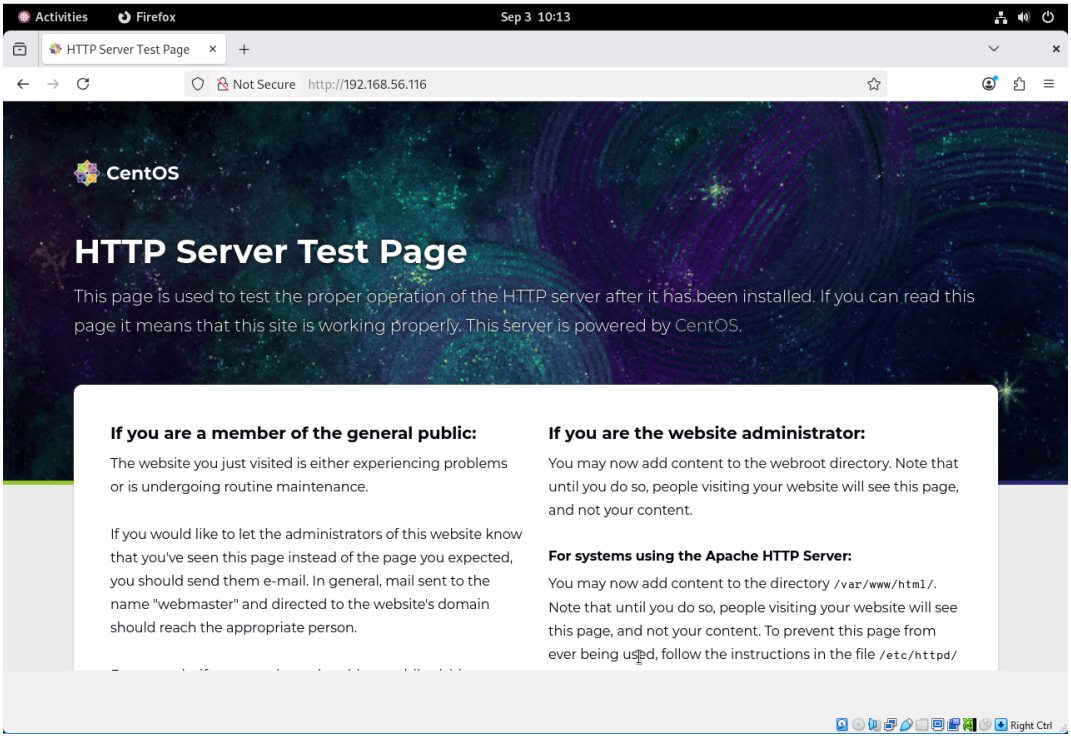
- name: "Mariadb-Restarting/Enabling"
  service:
    name: mariadb
    state: restarted
    enabled: true
```

This is because in CentOS, installed packages' services are not run automatically. Thus, we need to create the module to run it automatically.

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command *sudo systemctl stop httpd*. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server's IP address. You should not be getting a display because we stopped the httpd service already.

```
[momo@centos ~]$ sudo systemctl stop httpd
[sudo] password for momo:
[momo@centos ~]$
```

3. Go to the local machine and this time, run the *site.yml* file. Then after running the file, go again to the CentOS server and enter its IP address on the browser. Describe the result.



- After typing the IP Address, it redirected me to the Server Test Page, which indicated that it is working properly.

  To automatically enable the service every time we run the playbook, use the command *enabled: true* similar to Figure 7.1.2 and save the playbook.

```
- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:

1. **Creating a Role for Web Servers**
2. **Creating a Role for Database Servers**

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ls roles
db  web
momo1@workstation:~/Downloads/CPE212_MOMO$
```

3. **Combining Multiple Roles in a Playbook**

- **Web server role**

```
momo1@workstation:~/Downloads/CPE212_MOMO$ cat roles/web/tasks/main.yml
---
- name: Install Apache and PHP on Ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Install Apache and PHP on CentOS
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: Start and enable Apache on CentOS
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"
```

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --tags web --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Host '192.168.56.104' is using the discovered Python interpreter at '/usr/bin/python3.14', but future inst
allation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansibl
.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.104]
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin/python3.14', but future inst
allation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansibl
.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.103]
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.9', but future insta
llation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible
com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.102]

TASK [install updates (CentOS)] ************************************************************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injecte
d after the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/momo1/Downloads/CPE212_MOMO/site.yml:12:11

10       update_only: yes
11       update_cache: yes
12     when: ansible_distribution == "CentOS"
             ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [192.168.56.103]

PLAY RECAP *********************************************************************************************************
192.168.56.102             : ok=6    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=6    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.104             : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

momo1@workstation:~/Downloads/CPE212_MOMO$
```

- **Db server role**

```
momo1@workstation:~/Downloads/CPE212_MOMO$ cat roles/db/tasks/main.yml
---
- name: Install MariaDB on CentOS
  dnf:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: Start and enable MariaDB on CentOS
  service:
    name: mariadb
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"

- name: Install MariaDB on Ubuntu
  apt:
    name: mariadb-server
    state: latest
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: Start and enable MariaDB on Ubuntu
  service:
    name: mariadb
    state: started
    enabled: true
  when: ansible_distribution == "Ubuntu"
```

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --tags db --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin/python3.14', but future inst
allation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible
.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.103]
[WARNING]: Host '192.168.56.104' is using the discovered Python interpreter at '/usr/bin/python3.14', but future inst
allation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible
.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.104]
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.9', but future insta
llation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.
com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.102]

TASK [install updates (CentOS)] ************************************************************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injecte
d after the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/momo1/Downloads/CPE212_MOMO/site.yml:12:11

 10       update_only: yes
 11       update_cache: yes
 12     when: ansible_distribution == "CentOS"
              ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.

skipping: [192.168.56.103]
skipping: [192.168.56.104]
```

```
PLAY RECAP *********************************************************************************************************
192.168.56.102             : ok=6    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.103             : ok=7    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.104             : ok=3    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

momo1@workstation:~/Downloads/CPE212_MOMO$
```

- **Combined Roles**

```
momo1@workstation:~/Downloads/CPE212_MOMO$ cat site.yml
---

- hosts: all
  become: true
  pre_tasks:

  - name: install updates (CentOS)
    tags: always
    dnf:
      update_only: yes
      update_cache: yes
    when: ansible_distribution == "CentOS"

  - name: install updates (Ubuntu)
    tags: always
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  roles:
    - role: web
      tags:
        - web

- hosts: db_servers
  become: true
  roles:
    - role: db
      tags:
```

```
- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    tags: samba
    package:
      name: samba
      state: latest
momo1@workstation:~/Downloads/CPE212_MOMO$
```

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin/python3.14', but future inst
allation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible
.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.103]
[WARNING]: Host '192.168.56.104' is using the discovered Python interpreter at '/usr/bin/python3.14', but future inst
allation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible
.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.104]
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.9', but future insta
llation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.
com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.102]

TASK [install updates (CentOS)] ************************************************************************************
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: INJECT_FACTS_AS_VARS default to `True` is deprecated, top-level facts will not be auto injecte
d after the change. This feature will be removed from ansible-core version 2.24.
Origin: /home/momo1/Downloads/CPE212_MOMO/site.yml:12:11

10       update_only: yes
11       update_cache: yes
12     when: ansible_distribution == "CentOS"
             ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.
```

```
PLAY RECAP *********************************************************************************************************
192.168.56.102             : ok=8    changed=0    unreachable=0    failed=0    skipped=4    rescued=0    ignored=0
192.168.56.103             : ok=9    changed=1    unreachable=0    failed=0    skipped=5    rescued=0    ignored=0
192.168.56.104             : ok=4    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0

momo1@workstation:~/Downloads/CPE212_MOMO$
```

- **Tags**

```
momo1@workstation:~/Downloads/CPE212_MOMO$ ansible-playbook -i inventory.ini site.yml --list-tags

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [web]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [db]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
momo1@workstation:~/Downloads/CPE212_MOMO$
```

COMPUTER ENGINEERING

QUEZON CITY

REFLECTIONS

Reflections:

Answer the following:

1. What is the importance of putting our remote servers into groups?
   - Putting remote servers into groups makes it easier to manage servers with similar roles and it also allows us to target only the servers that need a specific task.
2. What is the importance of tags in playbooks?
   - Tags allow us to run only the specific tasks we need instead of running the whole playbook which makes managing the servers faster and more convenient.
3. Why do you think some services need to be managed automatically in playbooks?
   - Managing services automatically helps make sure they are running properly after the playbook is executed and it also saves time because we do not have to start or enable the services manually each time.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Understanding of Concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **The tool was only used to further understand some concepts. The tool was not used to generate answers and the codes were on my own and based on my own understanding.**

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